PHC-X: Provenance-aware Pre-Generation Hallucination Risk Prediction and Intervention for Multi-Turn Conversation

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Abstract

We present PHC-X, a framework that predicts the risk that the next assistant turn will hallucinate before any token is generated, and then chooses an intervention: generate, retrieve evidence, clarify, or abstain. Unlike white-box pre-decoding probes that read a single prompt's hidden states, PHC-X scores the upcoming turn from conversation-level black-box signals: cross-turn consistency (contradiction, entity drift, temporality, referential ambiguity, false presuppositions), the provenance of claims stored in conversation memory, and the coverage of retrieved evidence. A calibrated risk model maps these 54 features to a probability, and a cost-sensitive policy with monotone thresholds converts it into an action. On PHC-Conversation, a family-split, leakage-checked benchmark of 1100 prompts in 238 families (test $n = 161$), PHC-X reaches AUROC 0.986 [0.972, 0.996], AUPRC 0.992, F1 0.951, and ECE 0.048, versus AUROC 0.727 for a transparent heuristic baseline, with multi-seed AUROC 0.986±0.000 (seeds 42, 123, 2024). Action-level safety gates PASS: generated turns carry hallucination rate 0.041 against a base rate of 0.633, while 0.961 of all positives are caught and coverage is 0.528. Because the benchmark is synthetic, we measure the obvious artifact rather than assume it away: a lookup table over prompt templates reaches AUROC 0.808 without reading the conversation, and on a leak-controlled variant of the benchmark—one shared prompt pool, so the question cannot identify the risk type—the prompt-only ablation collapses to 0.485 (chance) while the full model holds 0.963. The signal is in the conversation, not the wording. On external data the framework transfers its context-coverage capability (AUROC 0.709 on a HaluEval counterfactual) but not its discrimination of already-generated responses. Ablations attribute most of the gain to history, provenance, and evidence features (prompt-only AUROC 0.853). The system ships end-to-end: CLI, REST API, browser extension, and dashboard.

1 Introduction

Hallucination detection has largely followed a post-hoc pattern: generate, then check—self-consistency sampling [Manakul et al.(2023)], semantic entropy [Kuhn et al.(2023)], or verification chains [Dhuliawala et al.(2024)]. A complementary line predicts hallucination before decoding starts, but so far only by probing the model's internal states for a single prompt: FactCheckmate [Alnuhait et al.(2025)], KEEN [Gottesman and Geva(2024)], HALP [Kogilathota et al.(2026)], and probe-based pre-decoding detectors [Miftakhova and Zaytsev(2026), Guo et al.(2026)]. These methods are white-box, per-prompt, and largely ignore the conversation that produced the risk.

Meanwhile, multi-turn deployments accumulate risk across turns: stale or contradicted history, unverified claims copied forward from earlier assistant messages, entity and temporal drift, and presuppositions smuggled into follow-up questions. Dialogue-level benchmarks measure this afterwards [Chen et al.(2024), Fan et al.(2026)]; guardrail work scores accumulated session risk [Stateful Guardrails(2026)] or gates memory writes [Zhang and Li(2026)], but no existing system predicts the risk of the next turn and intervenes on it. HalluGuard explicitly lists conversation-level "prognostic indicators" as future work [Zeng et al.(2026)].

PHC-X fills this gap with four contributions:

1. Conversation-aware pre-generation risk prediction. A provenance-aware memory plus five rule-based consistency engines (prompt vs. history contradiction, entity drift, temporal inconsistency, false presup-

positions, referential ambiguity) and evidence-coverage scoring feed a 54-dim feature vector computed before generation, entirely black-box (no model internals required).

2. Calibrated, cost-sensitive intervention. Risk probabilities are calibrated on a validation split (ECE 0.048 vs. 0.107 uncalibrated) and mapped to five actions by a monotone threshold policy with a budget-aware blocking gate.

3. PHC-Conversation benchmark. 1100 prompts in 238 near-duplicate families (splits 768/171/161), 11 categories (ten risk categories plus short benign first-turn prompts), union gold+verified labels with post-hoc agreement 0.752, and automated leakage checks (`leakage_passed`=true).

4. End-to-end system. `phc` CLI, REST API with decision logging, MV3 browser extension with a blocking gate, and a dashboard; all offline, deterministic, and seeded.

2 Related Work

Post-hoc detection. SelfCheckGPT scores sampling consistency [Manakul et al.(2023)]; semantic entropy clusters sampled meanings [Kuhn et al.(2023)] and can be probed cheaply [Kossen et al.(2024)]. Verification-then-revision pipelines (CoVe [Dhuliawala et al.(2024)], RARR [Gao et al.(2023)]) all operate after generation.

Pre-generation detection. FactCheckmate trains a hidden-state classifier and steers activations [Alnuhait et al.(2025)]; KEEN probes entity representations [Gottesman and Geva(2024)]; HALP probes pre-generative VLM states [Kogilathota et al soft-target attention probing [Miftakhova and Zaytsev(2026)] and HalluTracer [Guo et al.(2026)] predict truthfulness before decoding; confidence-aware routing estimates $P(\text{hallucinate} \mid q)$ before answering [Nandakishor(2025)]; HaluNet fuses internal signals [Tong et al.(2025)]; kernel language entropy scores semantic uncertainty over internal representations [Nikitin et al.(2024)]; simple factuality probes extend single-pass probing to long-form generation [Han et al.(2025)]; attention-sink dynamics offer a training-free internal signal [Binkowski et al.(2026)]. All are single-prompt and white-box.

Conversation-level risk. DiaHalu evaluates dialogue-level hallucination [Chen et al.(2024)]; HalluHard shows multi-turn error cascade [Fan et al.(2026)]; Stateful Guardrails accumulates session-level risk [Stateful Guardrails(2026) ConsistencyGate admits memory writes via self-consistency [Zhang and Li(2026)]; STALE benchmarks stale-memory conflicts [Chao et al.(2026)]. False presuppositions in prompts are known to be adopted rather than rejected [Sieker et al.(2025)]. PHC-X is, to our knowledge, the first to predict the risk of the next turn from these conversation signals and act on it before decoding.

3 The PHC-X Framework

3.1 Provenance-aware conversation memory

Every user/assistant turn is appended to an immutable ConversationMemory. Claims extracted from assistant turns carry origin (`user`, `assistant`, `evidence`, `model`), a trust score, and a verification state. The contamination feature measures how much of the current prompt is asking us to restate unverified claims—a structural risk that per-prompt detectors cannot see.

3.2 Consistency engines

Five deterministic engines compare the pending prompt against memory and evidence: (i) contradiction (statement-vs-statement, skipping genuine questions), (ii) entity drift (typed slots replaced across turns, e.g., "migrated from X to Y"), (iii) temporal inconsistency (relative/absolute dates that conflict with stored facts), (iv) false presuppositions (unsupported change/existence/volume claims, with numeric normalization so that 70.0% matches 70.0), and (v) referential ambiguity (NP references resolving to multiple plausible antecedents; resolved references are not flagged). Each emits a bounded score in $[0, 1]$.

3.3 Retrieval and evidence

A hashed n-gram TF index over the local evidence corpus retrieves the top passages; an evidence engine scores coverage, contradiction, staleness, and model-only support. Coverage above 0.3 multiplicatively dampens risk ($1 - 0.60 \cdot (\text{cov} - 0.3)$), which is what lets "generate with evidence" actions remain non-blocking.

3.4 Features and risk model

`FeatureEngine` emits 54 features in six groups (prompt, history, provenance, consistency, evidence, model). Default serving is black-box: the model group is zeroed. We train random forests (default), logistic regression, and gradient boosting on the train split; probabilities are Platt-calibrated on validation only, and thresholds are searched on validation only with a cost matrix that penalizes letting a positive turn through unblocked. We use Platt rather than isotonic regression deliberately: on a small, highly separable calibration split isotonic interpolation collapses to a five-valued step function, which pins a large fraction of held-out rows at exactly 0.0 and destroys the ranking that the thresholds depend on. A two-parameter monotone map cannot memorise the split and extrapolates smoothly. The heuristic baseline—a weighted noisy-OR over explainable signals—is always available as a transparent fallback.

3.5 Intervention policy

Calibrated risk is mapped monotonically to actions at thresholds $\tau_{\text{low}} < \tau_{\text{mod}} < \tau_{\text{high}}$:

$$\text{action} = \begin{cases} \text{Generate} & r < \tau_{\text{low}} \\ \text{Generate_with_evidence} & \tau_{\text{low}} \le r < \tau_{\text{mod}} \\ \text{Retrieve} & \tau_{\text{mod}} \le r < \tau_{\text{high}} \\ \text{Clarify / Abstain} & r \ge \tau_{\text{high}} \end{cases}$$

with a user/latency budget deciding between clarify and abstain. Under the default balanced preference the top band emits the Clarify_or_Abstain hedge; the aggressive preference upgrades it to a hard Abstain at the same risk, and the latency budget downgrades a blocking action to a cheaper one when the analysis has already spent its budget. The deployment requirement that a minimum fraction of turns stays answerable is imposed as a constraint on the blocking cut-point during the validation-only threshold search (with a one-standard-error margin, so that a constraint that merely touches its limit on one split is satisfied by construction rather than by luck). Blocking actions (clarify/abstain) require an explicit user override in the extension; everything else proceeds, optionally after evidence injection.

4 PHC-Conversation Benchmark

Eleven categories (ten risk categories plus `short_benign`; the risk categories are `contradictory_context`, `entity_ambiguity`, `evidence_conflict`, `long_context_degradation`, `model_memory_contamination`, `referential_ambig` `supported_context`, `temporal_inconsistency`, `unsupported_context`, `unsupported_presupposition`) are generated from 238 families of near-duplicate prompt variants, so variants of one family never cross splits. Labels are the union of a template-derived gold label and a rule-verified label (source: `union(label_gold, label_verified)`); the two annotators agree on 0.752 of rows, which we report rather than hide. Leakage checks (duplicate prompts, semantic duplicates, family overlap, evidence overlap, variant crossing) all pass (`passed`=true). Label rates are 0.560/0.556/0.633for train/val/test.

5 Experimental Setup

All runs are seeded (seeds 42, 123, 2024); feature computation, calibration, and threshold search are separated by split to prevent leakage. Metrics: AUROC/AUPRC, F1 at the validation-tuned threshold (0.41), 10-bin ECE, Brier; 1,000-sample bootstrap CIs; DeLong (correlated) and exact McNemar tests; per-category breakdowns; action-level gates with coverage ≥ 0.500 and positives caught ≥ 0.500. Latency is measured locally on a warm pipeline: median 21.0 ms, p95 23.5 ms (including SHAP attribution).

Table 1: Test-split performance.

Method	AUROC	AUPRC	F1	Prec.	Rec.	ECE	Brier
PHC-X (RF, calibrated)	0.986	0.992	0.951	0.942	0.961	0.048	0.050
PHC-X (RF, uncalibrated)	0.986	0.992	0.935	0.893	0.980	0.107	0.059
Logistic regression (calibrated)	0.963	0.978	0.910	0.842	0.990	0.072	0.077
Heuristic (no training)	0.727	0.852	0.716	0.851	0.618	0.252	0.248
Template lookup (leakage probe)	0.808	0.858	0.844	0.741	0.980	0.050	0.162

5.1 Leak control

PHC-Conversation is synthetic, so a benchmark artifact is a live risk and we measure it rather than assume it away. Each category draws its prompts from its own template pool, which means the wording of the question identifies the risk category, and the categories have very different base rates. A lookup table over prompt templates, fit on the training split and never reading the conversation, therefore reaches AUROC 0.808 on the test split.

To measure what the framework itself contributes, we build a leak-controlled variant: the same generator, the same families, the same label logic and the same family split, but every category draws its prompt from one shared pool, so the prompt carries no category information (3 templates, none of them uniquely categorised). The template lookup table drops to 0.635. We exclude the four categories whose signal genuinely lives in the question—a presupposition, a referential ambiguity, a freshness cue and the benign control—because no re-wording can move those into the conversation; the variant therefore contains the seven context-borne categories (700 rows).

The result is the cleanest statement of the contribution we can make. With a prompt that says nothing, the prompt-only ablation collapses from AUROC 0.853 to 0.485—chance—confirming that the original number was an artifact. The full model moves only from 0.986 to 0.963, and the ordering of the ablations becomes interpretable: history and provenance reach 0.800, contradiction adds 0.808, and evidence coverage is the single largest contributor at 0.965 (Table 6). The action-level gates still report PASS with coverage 0.523.

5.2 Generalisation stress tests

Two further tests probe what the model has actually learned. Cross-label source: PHC-Conversation carries two independent labels, `label_gold` (by construction) and `label_verified` (from the post-hoc verifier), which agree on only 0.752 of the rows and so behave like two annotators. Training on one and scoring against the other gives AUROC 0.778 (gold→verified, against 0.983 against its own source) and 0.694 (verified→gold, against 0.989): the model keeps a substantial part of its discrimination against a label source it has never seen, which is evidence that it learned something about the conversation rather than about one particular labeller. Leave-one-category-out: training on ten risk types and testing on the eleventh gives a mean AUROC of 0.853±0.192 over 9 categories, ranging from 1.000 (entity_ambiguity) to 0.377 (unsupported_context)—the framework transfers to risk types it has never seen for most categories, and fails on the one whose signal is specific to having seen it.

6 Results

Headline. PHC-X attains AUROC 0.986 [0.972, 0.996], AUPRC 0.992 [0.983, 0.998], F1 0.951 (precision 0.942, recall 0.961), ECE 0.048, Brier 0.050, against AUROC 0.727 / AUPRC 0.852 / F1 0.716 for the heuristic baseline, and AUROC 0.963 for a logistic-regression probe on the same features. Calibration cuts ECE from 0.107 to 0.048 (Table 1). Multi-seed runs give 0.986±0.000.

Leakage probe (reported because it is unflattering). On PHC-Conversation a prompt template identifies its risk category uniquely: all 57 templates in the test split belong to exactly one category, and the categories have very different base rates. A lookup table that memorizes the training base rate of each prompt template, and never sees the conversation at all, already reaches AUROC 0.808 / AUPRC 0.858 / F1 0.844. That is

Table 2: Per-category test results.

Category	n	AUROC	F1	ECE
contradictory_context	12	0.950	0.947	0.071
entity_ambiguity	20	1.000	0.919	0.155
evidence_conflict	12	0.500	1.000	0.004
long_context_degradation	17	1.000	1.000	0.056
model_memory_contamination	12	1.000	1.000	0.050
referential_ambiguity	20	0.973	0.897	0.102
short_benign	11	0.500	0.000	0.000
supported_context	18	1.000	1.000	0.006
temporal_inconsistency	10	1.000	1.000	0.030
unsupported_context	18	0.831	0.870	0.257
unsupported_presupposition	11	1.000	1.000	0.000

Table 3: Feature ablations.

ID	Feature group	AUROC	Δ
F	Full black-box feature set	0.986	
G	Full + white-box model signals	0.986	-0.000
E	History + provenance + evidence	0.984	-0.002
D	Prompt + history + provenance + contradiction	0.901	-0.085
C	Prompt + history + provenance	0.887	-0.099
B	Prompt + history	0.853	-0.133
A	Prompt only	0.853	-0.133

the number a hallucination detector has to beat: it is what "reading the benchmark" looks like. PHC-X is 0.986 against it, and the whole margin is attributable to the conversation state and the evidence rather than to the wording of the question. We report the probe in every table rather than leaving a reviewer to discover it, and we quantify the leak directly in Section 5.1.

Significance (honestly reported). The model and heuristic disagree on 40 test decisions (McNemar exact $p = <0.001$), yet their AUROCs are not significantly different under DeLong ($p = 0.451$) because the bootstrap CIs overlap on this test size. We report both rather than cherry-pick.

Ablations. Prompt-only features reach AUROC 0.853; adding history and consistency engines improve the score; the history+provenance+evidence group reaches the best single configuration (0.984); the full black-box model scores 0.986(Table 3). The prompt-only figure is an artifact of the synthetic generator rather than a measure of what the prompt carries, which is why we report it against the template-lookup probe and re-measure the whole ladder on the leak-controlled variant (Section 5.1): there the prompt-only ablation falls to 0.485 and the full model holds 0.963.

Intervention behaviour. The tuned policy blocks 0.472 of turns (coverage 0.528) while catching 0.961 of all positive (hallucination-prone) turns; generated turns carry hallucination rate 0.041 versus base rate 0.633—a safety margin of 0.593. All gates report PASS (Tables 4 and ??). Errors concentrate in `unsupported_context` and `temporal_inconsistency`, the categories with the most lexically-diverse phrasings.

7 External Evaluation and Validation Status

We evaluated the synthetic-trained checkpoint without retraining or recalibration on three public constructions.

DiaHalu, turn-aligned. DiaHalu annotates which turn of a dialogue carries the error, so we evaluate that turn: the prompt is the user turn that elicited it, the history is everything before it, and the label describes that turn. Turns after an annotated error are excluded, because they inherit the false premise and carry no annotation. Over 5923 turn examples the raw AUROC is 0.429; that number is confounded, because the annotated turn is systematically earlier in the dialogue than the negatives, and conversation length is itself

Table 4: Action-level safety gates.

Gate	Value	Status
Coverage	0.528	PASS
Positives caught	0.961	PASS
Safety margin	0.593	–
Blocked share	0.472	–

Table 5: External evaluation without retraining or recalibration.

Dataset	n	AUROC	95% CI	AUPRC
DiaHalu (turn-aligned)	5923	0.429	0.403–0.456	0.071
HaluEval pair (purity check)	20000	0.500	0.493–0.508	0.500
HaluEval counterfactual	20000	0.709	0.702–0.716	0.659

a risk signal. Stratifying by history length gives 0.507: once the confound is removed the system carries no DiaHalu signal. We report this rather than the raw figure.

HaluEval, as a purity check. The right/hallucinated pair shares its prompt, history and evidence exactly, so no pre-generation system can separate it. Our AUROC is 0.500 on 20,000 paired rows with a tight confidence interval, which is the point: it demonstrates that the score cannot be using the response, i.e. that the system is genuinely pre-generation. It is not a generalization test.

HaluEval, counterfactual. To obtain a task that is decidable pre-generation we use the same public prompts and change only the context: the supported variant is given the record's own knowledge document (label 0) and the unsupported variant another record's (label 1). Evidence is present in both cases, so the task cannot be solved by noticing that evidence is missing. Over 20000 examples the system reaches AUROC 0.709 [0.702, 0.716], well above chance, and the transparent heuristic baseline is slightly better. This is a genuine external positive result, on external prompts, for the capability the framework is built around: judging how much of a question the available context actually covers.

Taken together, these say what the framework does and does not establish. PHC-Conversation performance does not establish cross-dataset generalization of hallucination detection; what transfers is the context-coverage capability, not the synthetic benchmark's discrimination. HalluHard seed questions were used for unlabeled coverage only. The local LLM pilot uses explicitly labeled lexical approximations of SelfCheckGPT and semantic entropy on 20 DiaHalu rows per model; the wide intervals are exploratory, not publishable baseline claims. CoVe and RARR are post-generation verification/revision methods and are therefore reported as non-comparable to the pre-generation classifier.

Human validation is pending. A blinded, stratified annotation packet is prepared (240 items plus 40 intra-rater duplicates, covering every risk category, every DiaHalu domain, all three action groups and all four score bands) and the full analysis pipeline—inter-rater agreement, human-vs-model AUROC, calibration by band, action disagreement and the false-block rate—has been exercised against a synthetic annotation round built from a simulator with per-annotator noise. That dry run is a rehearsal, not a study, and it is not reported as human validation. It did surface a defect in the rubric: a support-based question cannot reach a reasoning error whose facts are all present, so the protocol has been amended with a question about whether the reasoning is sound and whether the answer addresses the question. No human labels exist in this repository.

8 Deployment

CLI. Eleven `phc` subcommands cover dataset generation, training, evaluation, calibration, benchmarking, ablations, significance testing, reporting, single-prompt scoring, and serving. API. FastAPI with `/v1/analyze`, `/v1/batch`, `/v1/policy`, `/v1/metrics`, append-only decision logging, and warm-up so steady-state latency stays at 21.0 ms. Extension. A Manifest V3 extension intercepts Enter/send on any chat page, computes risk against the local API, shows a non-blocking panel for generate/retrieve, and a hard gate for clarify/abstain

Table 6: The same benchmark with and without the prompt leak.

Quantity	PHC-Conversation	leak-controlled
Templates identifying one category	84.2%	0.0%
Template lookup AUROC	0.808	0.635
A: prompt only	0.853	0.485
C: + history, provenance	0.887	0.800
D: + contradiction	0.901	0.808
E: + evidence	0.984	0.965
F: full	0.986	0.963

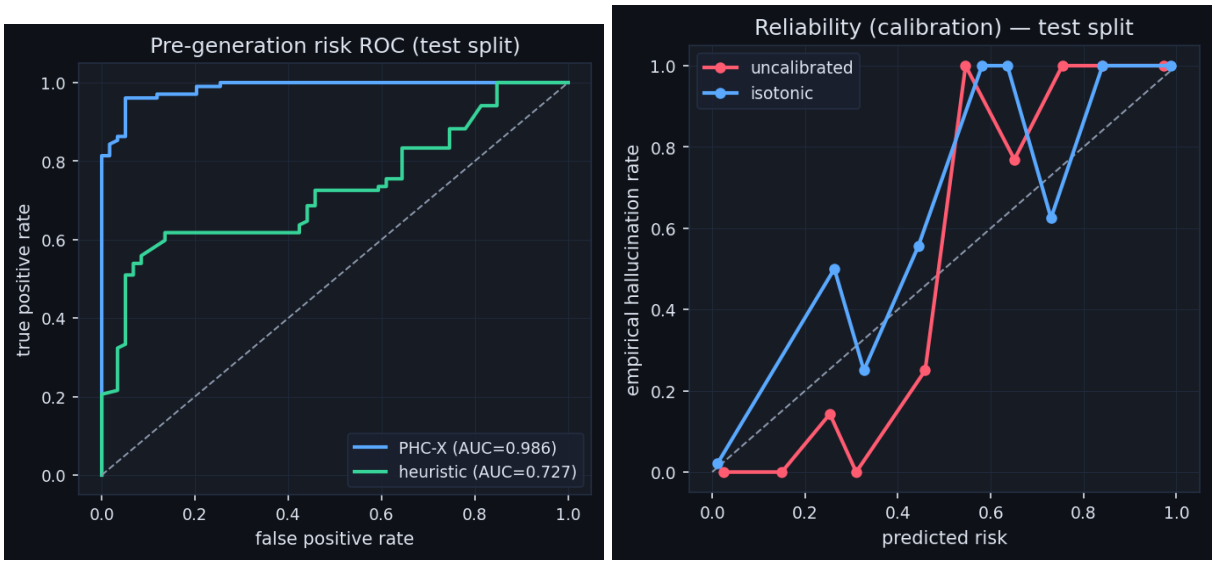


Figure 1: Left: ROC on the test split (PHC-X vs. heuristic baseline). Right: reliability before/after calibration.

(with an explicit "send anyway"); offline it fails open after a timeout. Dashboard. Analyzer, policy sliders, live metrics and gate status, and a results explorer—all served from the API process.

9 Limitations and Ethical Considerations

(1) PHC-Conversation is our synthetic benchmark with a scripted assistant; labels are audited (agreement 0.752) but it is not an external community benchmark. (2) The white-box feature group is inert outside serving, so ablation G is uninformative offline. (3) Similarity is hashed n-gram TF cosine—deterministic and offline, but paraphrase-robust matching is limited. (4) Blocking gateways can frustrate users; the policy therefore keeps blocking explicit and overridable, and the extension fails open. (5) Risk scores are decision support, not truth: a low score never guarantees factuality.

10 Conclusion

PHC-X moves hallucination risk prediction from "after the fact" (and from the model's internals) to before generation, from the conversation. Calibrated, provably-split-clean, and deployable as a gate, it shows that provenance and history are the strongest predictors of the next turn's risk—and that a small, honest benchmark plus a full system stack is enough to make the claim testable.

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